

MTE 544 Assignment 1

Armaan Sengupta
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1 Question 1: Mobile Manipulator

(a)

(b)

(c)

2 Question 2: Differential-Drive Simulation

(a) i. Constant velocities

1.

2.

3.

ii. Velocity profiles

(b)

3 Question 3: 3-Omnivheel Simulation

(a)

(b) I. Rotation inputs

II. Designed control signals

1. Straight line with a slope of 60 degrees
2. Two-metre diameter circle